

Figure6C

Emiliano Pereira

2025-08-06

Intro

Temperature range analysis of taxa between the Summer–Autumn and Winter–Spring semesters, to reproduce the results of Figure 6B from the original publication Seasonal dynamics of the coastal microbiome and its association with environmental factors.

1. Set the environment

```
library(tidyverse)
library(vegan)
```

2. Load data

asvs_workable.tsv contains the rarefied ASV abundance profiles, with samples as rows and ASVs as columns.

samo_metadata_workable.tsv contains the metadata for the samples. date2season2community.tsv is a table mapping the date, season, and community columns.

```
ABUND <- read_tsv("../data/asvs_workable.tsv.gz", show_col_types = FALSE)
METADATA <- read_tsv("../data/samo_metadata_workable.tsv", show_col_types = FALSE)
DATE2SEASON2COMMUNITY <- read_tsv("../data/date2season2community.tsv", show_col_types = FALSE)
```

3. Convert ABUND to long format and map metadata

```
ABUND_long <- ABUND %>%
  mutate(Date = as.Date(Date)) %>%
  pivot_longer(~Date, names_to = "asv_id", values_to = "abund") %>%
  left_join(x = .,
    y = METADATA %>% dplyr::select(Date, sample_name, Temperature_C),
    by = c("Date")) %>%
  left_join(x = .,
    y = DATE2SEASON2COMMUNITY ,
    by = "Date") %>%
  filter(abund > 0) # it is critical to remove zeros, other wise asv ranges in S1 and S2 are
```

4. Compute optimal temperature for each ASV

```
asvs2optimal_temp <- ABUND_long %>%
  group_by(asv_id) %>%
  mutate(asvs_optimal_temp = sum(abund*Temperature_C)/sum(abund)) %>%
  mutate(asvs_optimal_temp_ratio = asvs_optimal_temp/Temperature_C)
```

5. Create boxplot comparing Semester's temperature vs ASVs optimal temperature

```
text_size <- 12

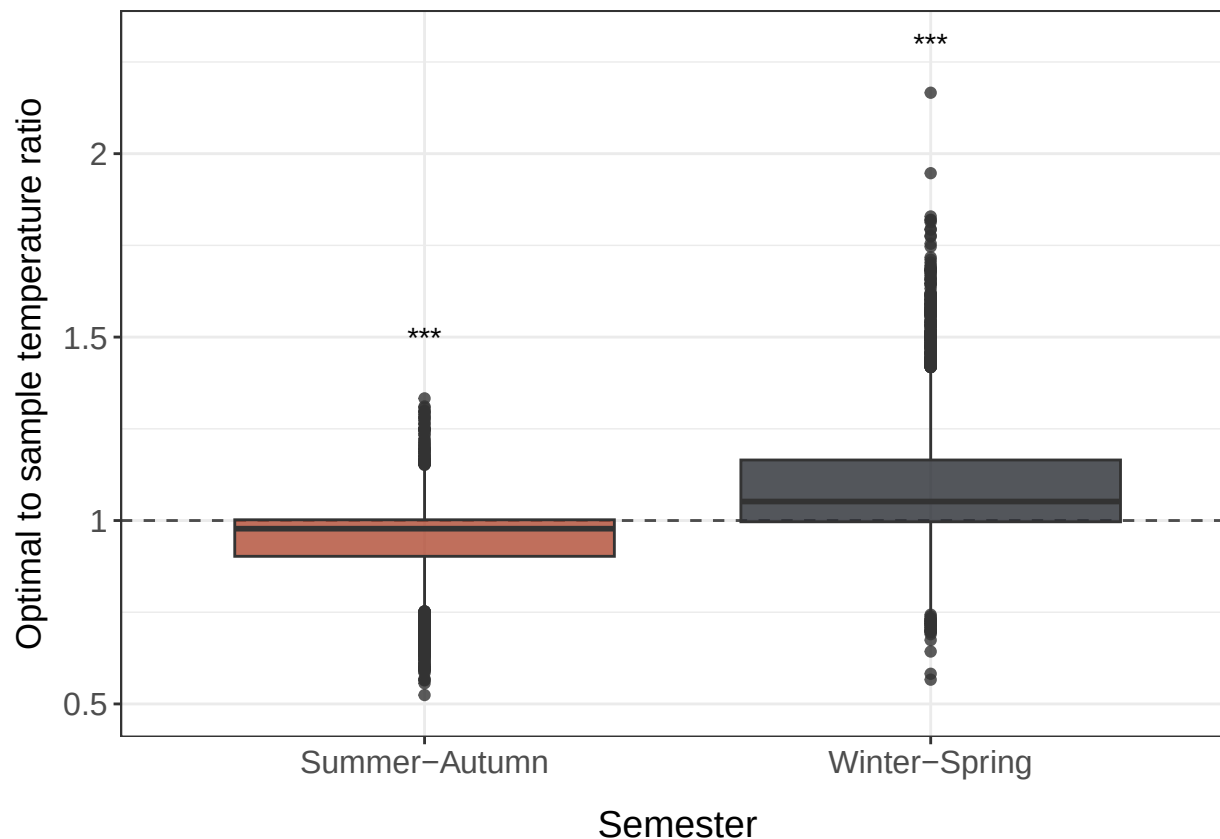
opt_temp_boxplot <- ggplot(asvs2optimal_temp, aes(x = Community,
                                                  y = asvs_optimal_temp_ratio, fill = Community)) +
  # add a line at y = 1
  geom_hline(yintercept = 1, linetype = "dashed", color = "gray30") +
  geom_boxplot(alpha = 0.8) +
  theme_minimal() +
  xlab("Semester") +
  ylab("Optimal to sample temperature ratio") +
  scale_fill_manual(values = c("#b14b34", "#292c33"),
                    labels = c("S1" = "Summer-Autumn", "S2" = "Winter-Spring"),
                    name = "Semester") +
  scale_x_discrete(labels = c("S1" = "Summer-Autumn", "S2" = "Winter-Spring")) +
  # set breaks in y axis

  scale_y_continuous(breaks = seq(0.5, 2, 0.5),
                    labels = seq(0.5, 2, 0.5),
                    limits = c(0.5, 2.3)) +

  theme_bw() +
  theme(
    axis.text.x = element_text(size = text_size),
    axis.text.y = element_text(size = text_size),
    axis.title.x = element_text(size = text_size + 2, margin = unit(c(4, 0, 0, 0), "mm")),
    axis.title.y = element_text(size = text_size + 1, margin = unit(c(0, 2, 0, 0), "mm")),
    strip.text = element_text(size = text_size + 4),
    strip.background = element_blank(),
    # increase legend text
    legend.text = element_text(size = text_size),
    legend.title = element_text(size = text_size + 4)
  ) +
  # remove legend
  theme(legend.position = "none")

opt_temp_boxplot_wtext <- opt_temp_boxplot +
  # add text to the plot
  annotate("text", x = 2, y = 2.3, label = "***", size = 4) +
  annotate("text", x = 1, y = 1.5, label = "***", size = 4)

opt_temp_boxplot_wtext
```



6. Run Wilcoxon Signed Rank Sum test: Summer-Autumn lower than 1

```
asvs_optimal_temp_ratio_summer_autumn <- asvs2optimal_temp %>%
  filter(Community == "S1") %>%
  pull(asvs_optimal_temp_ratio)

wtest_summer_autumn <- wilcox.test(asvs_optimal_temp_ratio_summer_autumn,
  mu = 1, alternative = "less")

wtest_summer_autumn
```

```
##
## Wilcoxon signed rank test with continuity correction
##
## data: asvs_optimal_temp_ratio_summer_autumn
## V = 2748838, p-value < 2.2e-16
## alternative hypothesis: true location is less than 1
```

7. Run Wilcoxon Signed Rank Sum test: Winter-Spring greater than 1

```
asvs_optimal_temp_ratio_winter_spring <- asvs2optimal_temp %>%
  filter(Community == "S2") %>%
  pull(asvs_optimal_temp_ratio)

wtest_winter_spring <- wilcox.test(asvs_optimal_temp_ratio_winter_spring,
  mu = 1, alternative = "greater")
```

```
wtest_winter_spring
```

```
##
## Wilcoxon signed rank test with continuity correction
##
## data:  asvs_optimal_temp_ratio_winter_spring
## V = 9540562, p-value < 2.2e-16
## alternative hypothesis: true location is greater than 1
```

8. Print session info

```
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 20.04.6 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.9.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.9.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C              LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
##  [6] LC_MESSAGES=en_US.UTF-8  LC_PAPER=en_US.UTF-8     LC_NAME=C                LC_ADDRESS=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
##  [1] fANCOVA_0.6-1      maps_3.4.2          corrgram_1.14       ggpubr_0.6.0        ggcorrplot_0.1.4.1
##  [8] permute_0.9-7      lubridate_1.9.3     forcats_1.0.0       stringr_1.5.1       dplyr_1.1.4
## [15] tidyr_1.3.1        tibble_3.2.1        ggplot2_3.5.1       tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.5      xfun_0.48           rstatix_0.7.2       tzdb_0.4.0          vctrs_0.6.5         tools_
##  [9] fansi_1.0.6       highr_0.11          cluster_2.1.6       pkgconfig_2.0.3     Matrix_1.7-0        lifecycle_
## [17] tinytex_0.53      munsell_0.5.1       carData_3.0-5       htmltools_0.5.8.1   yaml_2.3.10         Formula_
## [25] car_3.1-3         MASS_7.3-61         abind_1.4-8         nlme_3.1-166        tidyselect_1.2.1     digest_
## [33] splines_4.4.2     cowplot_1.1.3       fastmap_1.2.0       colorspace_2.1-1    cli_3.6.3           magrittr_
## [41] withr_3.0.1       scales_1.3.0        backports_1.5.0     bit64_4.5.2         timechange_0.3.0    rmarkdown_
## [49] hms_1.1.3         evaluate_1.0.1      knitr_1.48          mgcv_1.9-1          rlang_1.1.4         glue_
## [57] R6_2.5.1
```